

Solutions of Problems on CH 26

3. (a) How much charge is on each plate of a $4.00\text{-}\mu\text{F}$ capacitor when it is connected to a 12.0-V battery?
(b) If this same capacitor is connected to a 1.50-V battery, what charge is stored?

(a) $Q = C\Delta V = (4.00 \times 10^{-6} \text{ F})(12.0 \text{ V}) = 4.80 \times 10^{-5} \text{ C} = \boxed{48.0 \mu\text{C}}$

(b) $Q = C\Delta V = (4.00 \times 10^{-6} \text{ F})(1.50 \text{ V}) = 6.00 \times 10^{-6} \text{ C} = \boxed{6.00 \mu\text{C}}$

4. An air-filled spherical capacitor is constructed with inner- and outer-shell radii of 7.00 cm and 14.0 cm, respectively. (a) Calculate the capacitance of the device. (b) What potential difference between the spheres results in a 4.00- μC charge on the capacitor?

(a) For a spherical capacitor with inner radius a and outer radius b ,

$$C = \frac{ab}{k_e(b-a)} = \frac{(0.0700 \text{ m})(0.140 \text{ m})}{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(0.140 \text{ m} - 0.0700 \text{ m})}$$
$$= \boxed{15.6 \text{ pF}}$$

(b) $\Delta V = \frac{Q}{C} = \frac{4.00 \times 10^{-6} \text{ C}}{1.56 \times 10^{-11} \text{ F}} = 2.57 \times 10^5 \text{ V} = \boxed{257 \text{ kV}}$

5. A 50.0-m length of coaxial cable has an inner conductor that has a diameter of 2.58 mm and carries a charge of $8.10 \mu\text{C}$. The surrounding conductor has an inner diameter of 7.27 mm and a charge of $-8.10 \mu\text{C}$. Assume the region between the conductors is air. (a) What is the capacitance of this cable? (b) What is the potential difference between the two conductors?

(a) The capacitance of a cylindrical capacitor is

$$\begin{aligned}C &= \frac{\ell}{2k_e \ln(b/a)} \\&= \frac{50.0 \text{ m}}{2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \ln(7.27 \text{ mm}/2.58 \text{ mm})} \\&= \boxed{2.68 \text{ nF}}\end{aligned}$$

(b) Method 1: $\Delta V = 2k_e \lambda \ln\left(\frac{b}{a}\right)$

$$\lambda = \frac{Q}{\ell} = \frac{8.10 \times 10^{-6} \text{ C}}{50.0 \text{ m}} = 1.62 \times 10^{-7} \text{ C/m}$$

$$\begin{aligned}\Delta V &= 2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.62 \times 10^{-7} \text{ C/m}) \ln\left(\frac{7.27 \text{ mm}}{2.58 \text{ mm}}\right) \\&= \boxed{3.02 \text{ kV}}\end{aligned}$$

$$\text{Method 2: } \Delta V = \frac{Q}{C} = \frac{8.10 \times 10^{-6} \text{ C}}{2.68 \times 10^{-9} \text{ F}} = \boxed{3.02 \text{ kV}}$$

7. When a potential difference of 150 V is applied to the plates of a parallel-plate capacitor, the plates carry a surface charge density of 30.0 nC/cm². What is the spacing between the plates?

We have $Q = C\Delta V$ and $C = \epsilon_0 A / d$. Thus, $Q = \epsilon_0 A \Delta V / d$

The surface charge density on each plate has the same magnitude, given by

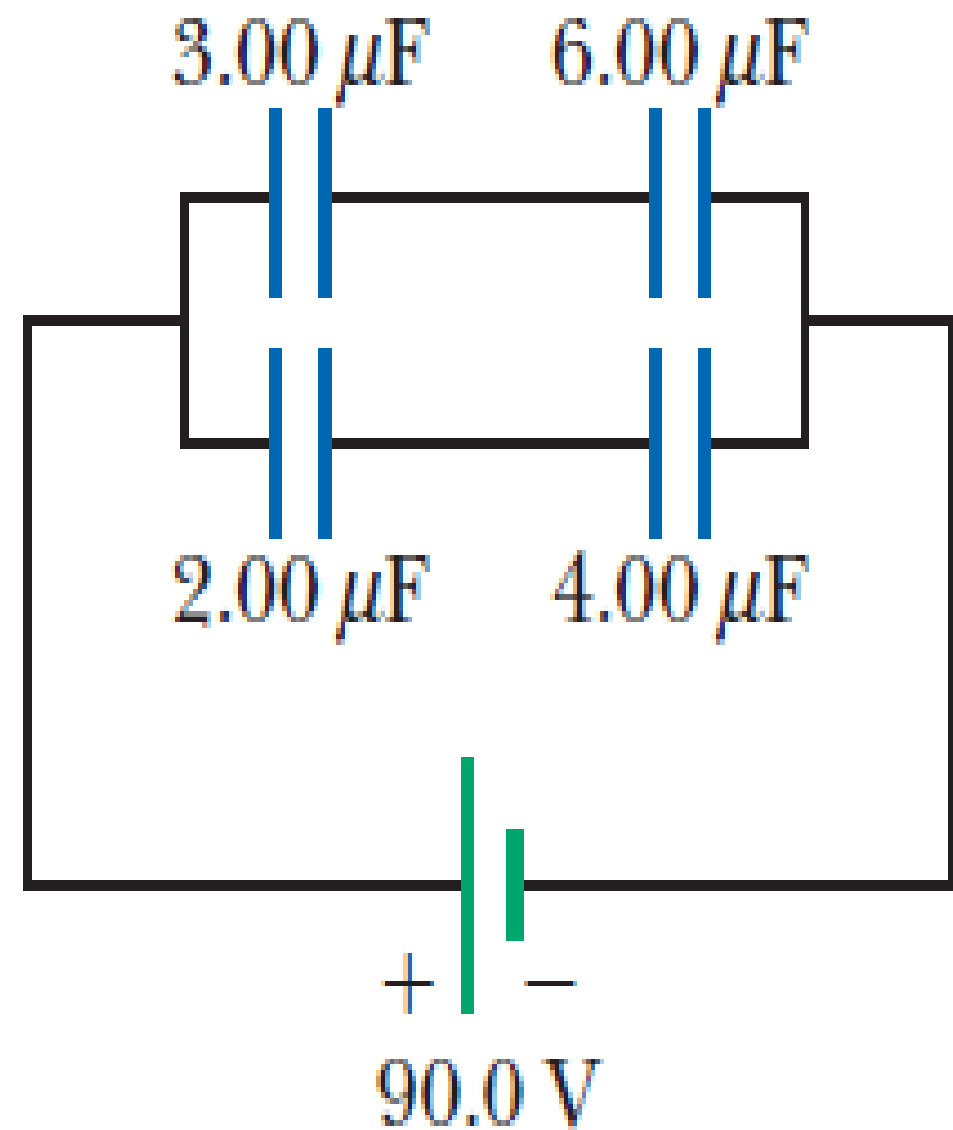
$$\sigma = \frac{Q}{A} = \frac{\epsilon_0 \Delta V}{d}$$

Thus,

$$d = \frac{\epsilon_0 \Delta V}{Q/A} = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(150 \text{ V})}{(30.0 \times 10^{-9} \text{ C}/\text{cm}^2)}$$

$$d = \left(4.43 \times 10^{-2} \frac{\text{V} \cdot \text{C} \cdot \text{cm}^2}{\text{N} \cdot \text{m}^2} \right) \frac{(1 \text{ m}^2)}{(10^4 \text{ cm}^2)} \frac{\text{J}}{\text{V} \cdot \text{C}} \frac{\text{N} \cdot \text{m}}{\text{J}} = \boxed{4.43 \text{ } \mu\text{m}}$$

19. For the system of four capacitors shown in Figure P26.19, find (a) the equivalent capacitance of the system, (b) the charge on each capacitor, and (c) the potential difference across each capacitor.



- (a) The equivalent capacitance of the series combination in the upper branch is

$$\frac{1}{C_{\text{upper}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{3.00 \mu\text{F}} + \frac{1}{6.00 \mu\text{F}} \rightarrow C_{\text{upper}} = 2.00 \mu\text{F}$$

Likewise, the equivalent capacitance of the series combination in the lower branch is

$$\frac{1}{C_{\text{lower}}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2.00 \mu\text{F}} + \frac{1}{4.00 \mu\text{F}} \rightarrow C_{\text{lower}} = 1.33 \mu\text{F}$$

These two equivalent capacitances are connected in parallel with each other, so the equivalent capacitance for the entire circuit is

$$C_{\text{eq}} = C_{\text{upper}} + C_{\text{lower}} = 2.00 \mu\text{F} + 1.33 \mu\text{F} = \boxed{3.33 \mu\text{F}}$$

(b) Note that the same potential difference, equal to the potential difference of the battery, exists across both the upper and lower branches. Each of the capacitors in series combination holds the same charge as that on the equivalent capacitor. For the upper branch:

$$Q_3 = Q_6 = Q_{\text{upper}} = C_{\text{upper}} (\Delta V) = (2.00 \mu\text{F})(90.0 \text{ V}) = 180 \mu\text{C}$$

so, $180 \mu\text{C}$ on the $3.00\text{-}\mu\text{F}$ and the $6.00\text{-}\mu\text{F}$ capacitors

For the lower branch:

$$Q_2 = Q_4 = Q_{\text{lower}} = C_{\text{lower}} (\Delta V) = (1.33 \mu\text{F})(90.0 \text{ V}) = 120 \mu\text{C}$$

so, $120 \mu\text{C}$ on the $2.00\text{-}\mu\text{F}$ and $4.00\text{-}\mu\text{F}$ capacitors

(c) The potential difference across each of the capacitors in the circuit is:

$$\Delta V_2 = \frac{Q_2}{C_2} = \frac{120 \mu\text{C}}{2.00 \mu\text{F}} = \boxed{60.0 \text{ V}}$$

$$\Delta V_3 = \frac{Q_3}{C_3} = \frac{180 \mu\text{C}}{3.00 \mu\text{F}} = \boxed{60.0 \text{ V}}$$

$\boxed{60.0 \text{ V across the } 3.00\text{-}\mu\text{F} \text{ and the } 2.00\text{-}\mu\text{F} \text{ capacitors}}$

$$\Delta V_4 = \frac{Q_4}{C_4} = \frac{120 \mu\text{C}}{4.00 \mu\text{F}} = \boxed{30.0 \text{ V}}$$

$$\Delta V_6 = \frac{Q_6}{C_6} = \frac{180 \mu\text{C}}{6.00 \mu\text{F}} = \boxed{30.0 \text{ V}}$$

$\boxed{30.0 \text{ V across the } 6.00\text{-}\mu\text{F} \text{ and the } 4.00\text{-}\mu\text{F} \text{ capacitors}}$

20. Three capacitors are connected to a battery as shown in Figure P26.20. Their capacitances are $C_1 = 3C$, $C_2 = C$, and $C_3 = 5C$. (a) What is the equivalent capacitance of this set of capacitors? (b) State the ranking of the capacitors according to the charge they store from largest to smallest. (c) Rank the capacitors according to the potential differences across them from largest to smallest. (d) **What If?** Assume C_3 is increased. Explain what happens to the charge stored by each capacitor.

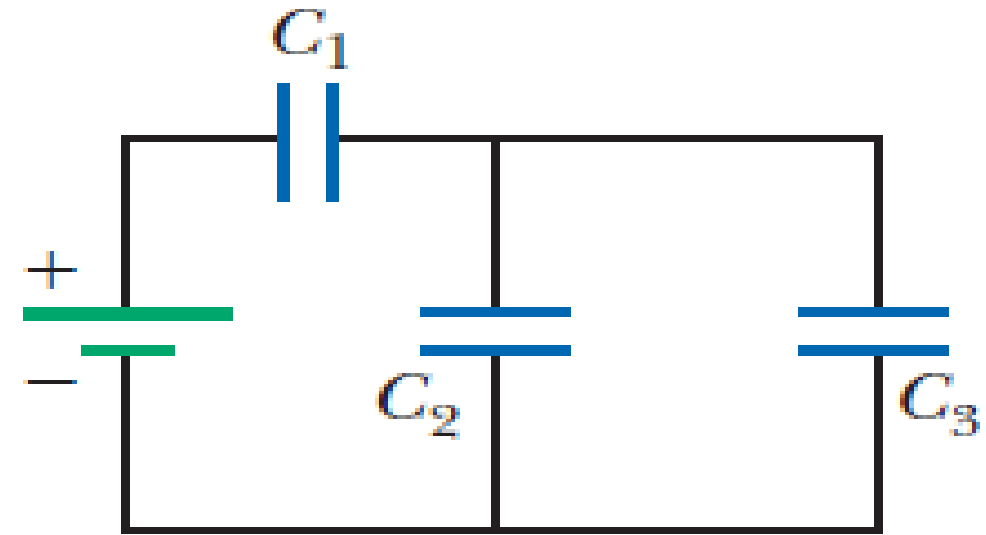


Figure P26.20

(a) Capacitors 2 and 3 are in parallel and present equivalent capacitance $6C$. This is in series with capacitor 1, so the battery sees capacitance $\left[\frac{1}{3C} + \frac{1}{6C} \right]^{-1} = \boxed{2C}$.

(b) If they were initially uncharged, C_1 stores the same charge as C_2 and C_3 together. With greater capacitance, C_3 stores more charge than C_2 . Then $\boxed{Q_1 > Q_3 > Q_2}$.

(c) The $(C_2 \parallel C_3)$ equivalent capacitor stores the same charge as C_1 . Since it has greater capacitance, $\Delta V = \frac{Q}{C}$ implies that it has smaller potential difference across it than C_1 . In parallel with each other, C_2 and C_3 have equal voltages: $\boxed{\Delta V_1 > \Delta V_2 = \Delta V_3}$.

(d) If C_3 is increased, the overall equivalent capacitance increases. More charge moves through the battery and Q increases. As ΔV_1 increases, ΔV_2 must decrease so Q_2 decreases. Then Q_3 must increase even more: Q_3 and Q_1 increase; Q_2 decreases.

32. (a) A $3.00\text{-}\mu\text{F}$ capacitor is connected to a 12.0-V battery. **W** How much energy is stored in the capacitor? (b) Had the capacitor been connected to a 6.00-V battery, how much energy would have been stored?

$$(a) \quad U_E = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}(3.00 \mu\text{F})(12.0 \text{ V})^2 = \boxed{216 \mu\text{J}}$$

$$(b) \quad U_E = \frac{1}{2}C(\Delta V)^2 = \frac{1}{2}(3.00 \mu\text{F})(6.00 \text{ V})^2 = \boxed{54.0 \mu\text{J}}$$

44. The voltage across an air-filled parallel-plate capacitor is measured to be 85.0 V. When a dielectric is inserted and completely fills the space between the plates as in Figure P26.44, the voltage drops to 25.0 V. (a) What is the dielectric constant of the inserted material? (b) Can you identify the dielectric? If so, what is it? (c) If the dielectric does not completely fill the space between the plates, what could you conclude about the voltage across the plates?

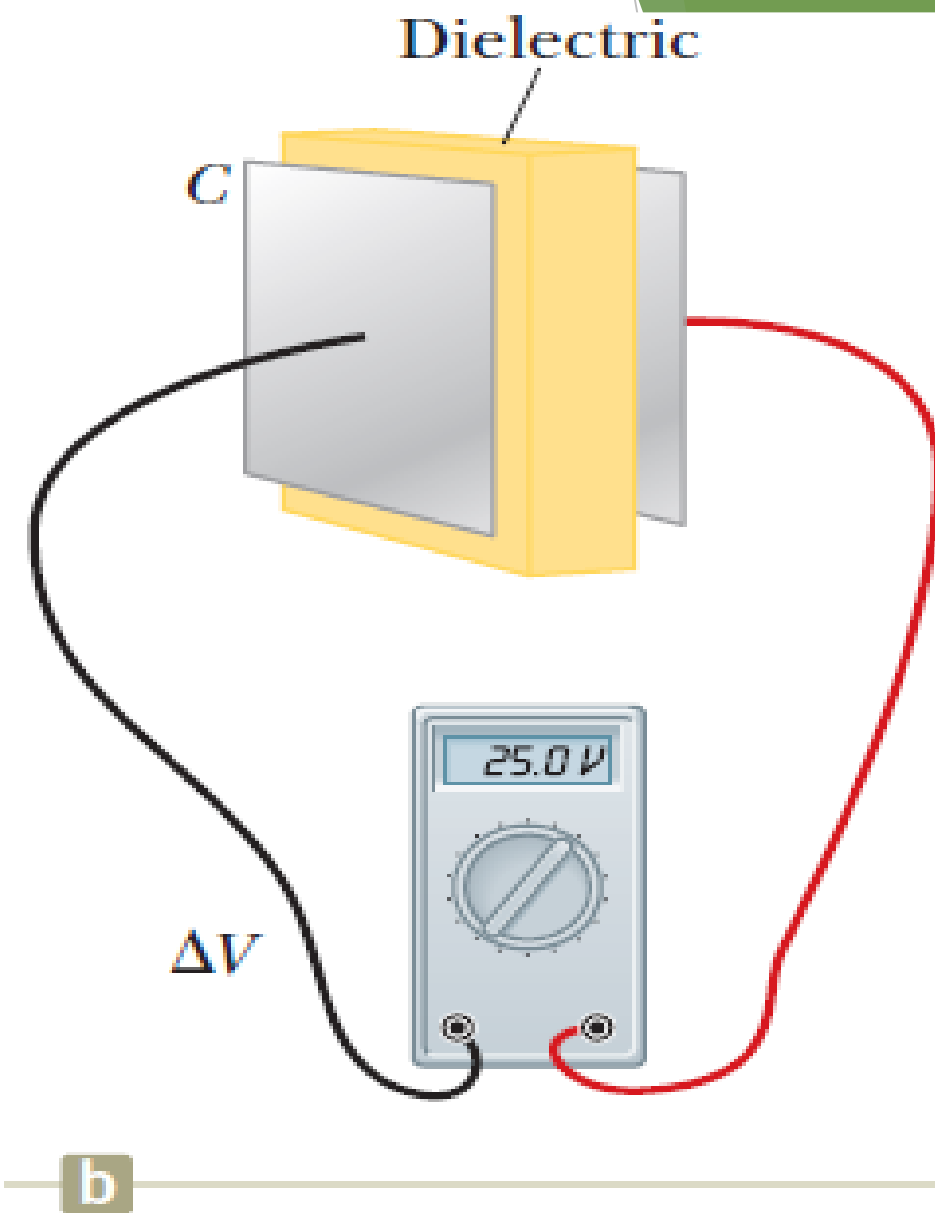
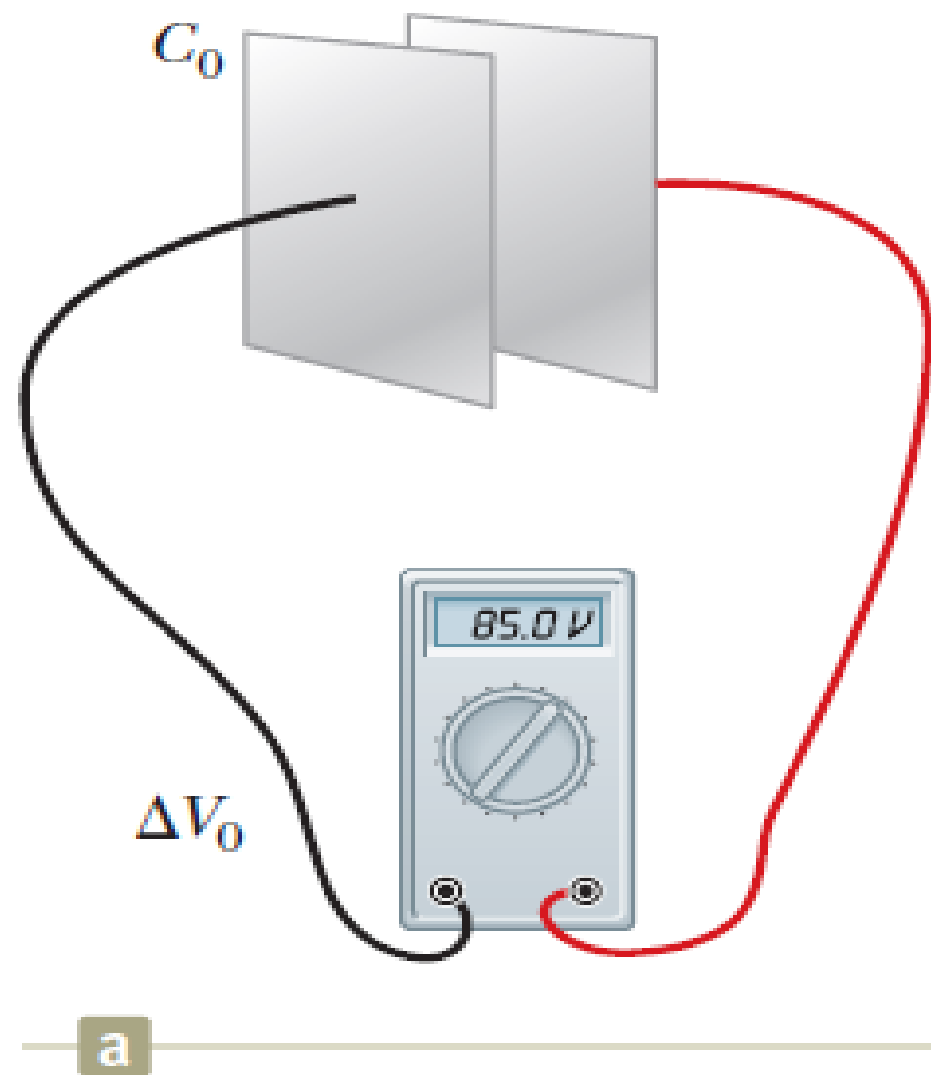


Figure P26.44

- (a) Note that the charge on the plates remains constant at the original value, Q_0 , as the dielectric is inserted. Thus, the change in the potential difference, $\Delta V = Q / C$, is due to a change in capacitance alone. The ratio of the final and initial capacitances is

$$\frac{C_f}{C_i} = \frac{\kappa \epsilon_0 A/d}{\epsilon_0 A/d} = \kappa$$

and
$$\frac{C_f}{C_i} = \frac{Q_0/(\Delta V)_f}{Q_0/(\Delta V)_i} = \frac{(\Delta V)_i}{(\Delta V)_f} = \frac{85.0 \text{ V}}{25.0 \text{ V}} = 3.40$$

Thus, the dielectric constant of the inserted material is $\boxed{\kappa = 3.40}$.

- (b) The material is probably $\boxed{\text{nylon}}$ (see Table 26.1).

(c) The presence of a dielectric weakens the field between plates, and the weaker field, for the same charge on the plates, results in a smaller potential difference. If the dielectric only partially filled the space between the plates, the field is weakened only within the dielectric and not in the remaining air-filled space, so the potential difference would not be as small. The voltage would lie somewhere between 25.0 V and 85.0 V.

45. Determine (a) the capacitance and (b) the maximum **W** potential difference that can be applied to a Teflon-filled parallel-plate capacitor having a plate area of 1.75 cm^2 and a plate separation of 0.0400 mm .

$$\begin{aligned} \text{(a)} \quad C &= \frac{\kappa \epsilon_0 A}{d} = \frac{2.10(8.85 \times 10^{-12} \text{ F/m})(1.75 \times 10^{-4} \text{ m}^2)}{4.00 \times 10^{-5} \text{ m}} = 8.13 \times 10^{-11} \text{ F} \\ &= \boxed{81.3 \text{ pF}} \end{aligned}$$

$$\text{(b)} \quad \Delta V_{\text{max}} = E_{\text{max}} d = (60.0 \times 10^6 \text{ V/m})(4.00 \times 10^{-5} \text{ m}) = \boxed{2.40 \text{ kV}}$$

47. A parallel-plate capacitor in air has a plate separation of 1.50 cm and a plate area of 25.0 cm^2 . The plates are charged to a potential difference of 250 V and disconnected from the source. The capacitor is then immersed in distilled water. Assume the liquid is an insulator. Determine (a) the charge on the plates before and after immersion, (b) the capacitance and potential difference after immersion, and (c) the change in energy of the capacitor.

(a) The charge is the same before and after immersion, with value

$$Q = C_i (\Delta V)_i = \frac{\epsilon_0 A (\Delta V)_i}{d}$$

$$Q = \frac{(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(25.0 \times 10^{-4} \text{ m}^2)(250 \text{ V})}{1.50 \times 10^{-2} \text{ m}} = \boxed{369 \text{ pC}}$$

(b) Finally,

$$C_f = \frac{\kappa \epsilon_0 A}{d} = \frac{Q}{(\Delta V)_f}:$$

$$C_f = \frac{(80)(8.85 \times 10^{-12} \text{ C}^2 / \text{N} \cdot \text{m}^2)(25.0 \times 10^{-4} \text{ m}^2)}{1.50 \times 10^{-2} \text{ m}} = \boxed{1.20 \times 10^{-10} \text{ F}}$$

$$\begin{aligned} \text{and } (\Delta V)_f &= \frac{Q}{C_f} = \frac{C_i (\Delta V)_i}{C_f} = \frac{(\epsilon_0 A / d)}{(\kappa \epsilon_0 A / d)} (\Delta V)_i = \frac{(\Delta V)_i}{\kappa} = \frac{250 \text{ V}}{80} \\ &= \boxed{3.10 \text{ V}} \end{aligned}$$

(c) Originally, $U_i = \frac{1}{2}C_i(\Delta V)_i^2 = \frac{\epsilon_0 A(\Delta V)_i^2}{2d}$.

Finally, $U_f = \frac{1}{2}C_f(\Delta V)_f^2 = \frac{1}{2}\left(\frac{\kappa\epsilon_0 A}{d}\right)\left(\frac{(\Delta V)_i}{\kappa}\right)^2 = \frac{\epsilon_0 A(\Delta V)_i^2}{2d\kappa},$

where, from Table 26.1, $\kappa = 80$ for distilled water. So,

$$\begin{aligned}\Delta U &= U_f - U_i = \frac{\epsilon_0 A(\Delta V)_i^2}{2d\kappa} - \frac{\epsilon_0 A(\Delta V)_i^2}{2d} = \frac{\epsilon_0 A(\Delta V)_i^2}{2d}\left(\frac{1}{\kappa} - 1\right) \\ &= \frac{\epsilon_0 A(\Delta V)_i^2(1 - \kappa)}{2d\kappa} \\ \Delta U &= \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(25.0 \times 10^{-4} \text{ m}^2)(250 \text{ V})^2(1 - 80)}{2(1.50 \times 10^{-2} \text{ m})(80)} \\ &= -4.55 \times 10^{-8} \text{ J} = \boxed{-45.5 \text{ nJ}}\end{aligned}$$